

# Evaluating Digital Demand Signals and Territorial Mobility Patterns to Support Airline Seat Allocation

*International Outbound Tourism Analysis in Spain (2019-2025)*

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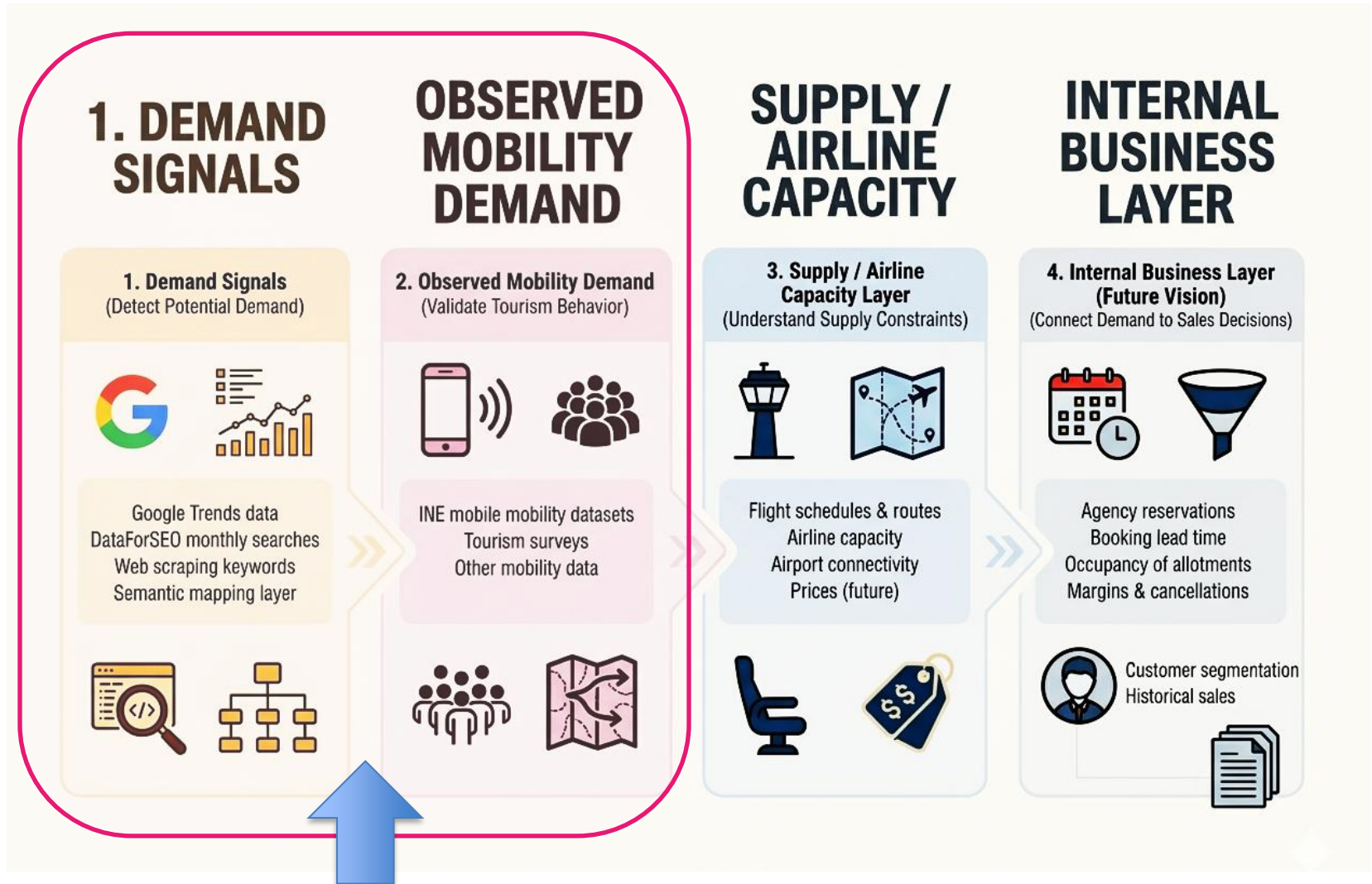
# THE BUSINESS PROBLEM

## **Contractual and Financial Risks of Tour operators and Travel Agencies:**

- Flight allotments decided 11-12 months in advance.
- They imply high deposits and other contractual risks
- Unsold seats create financial losses.

**Can digital demand signals anticipate  
outbound tourism demand to support  
better airline seat allocation decisions?**

# COMPLETE PROJECT FRAMEWORK



# DATA SOURCES & METHODOLOGY



## INE Mobility

Experimental statistics on outbound mobility (2019-2025) derived from **mobile phone signaling**.

- ~75% coverage of the Spanish market.
- Granularity: Province-Country-Month.
- Used as the **ground truth** for real demand.



## Digital Signals

Analysis of search intent and micro-dynamics via global search telemetry.

- **DataForSEO**: Absolute monthly search volumes (high stability).
- **Google Trends**: Relative search interest indices.
- Mapped through a 12-month **lag analysis**.



## Intelligence Layer

Semantic enrichment and domain curation to transform raw data into operation intelligence.

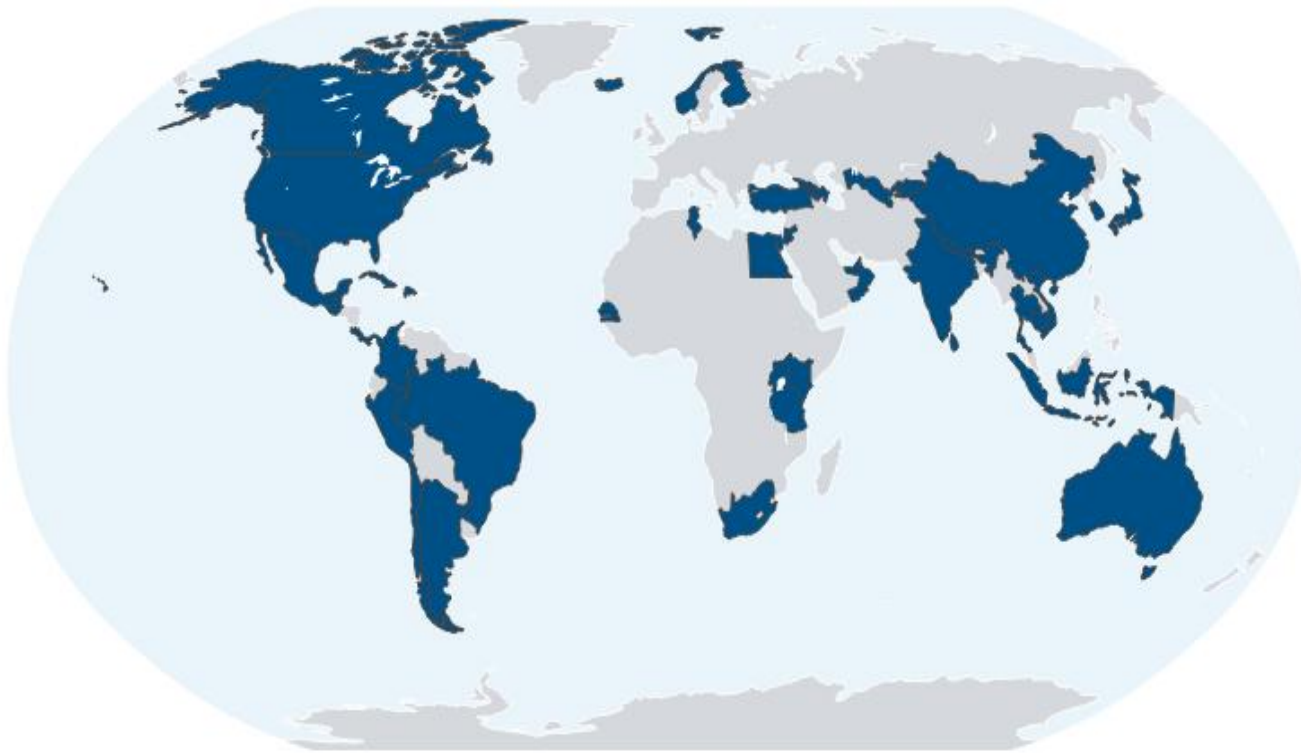
- **Web Scraping**: Commercial keywords from sector-leading websites.
- **Semantic Mapping**: Curation of multi-country regions (Balkans, Indochina).
- **Geospatial Data**: CCAA and Province integration.

**Observed demand**

**Google searches  
(absolute & relative)**

**Keyword generation**

## ANALYTICAL SCOPE: long-haul, agency-mediated destinations



■ Not in scope — core Europe, border mobility, VFR demand

### Why exclude Europe?

Among leisure travelers, **45%** of trips to América use travel packages vs only **18%** for Europe.

The “**viaje a**” search signal aligns with agency-mediated demand, where package rates are **2–2.5x** higher than Europe.

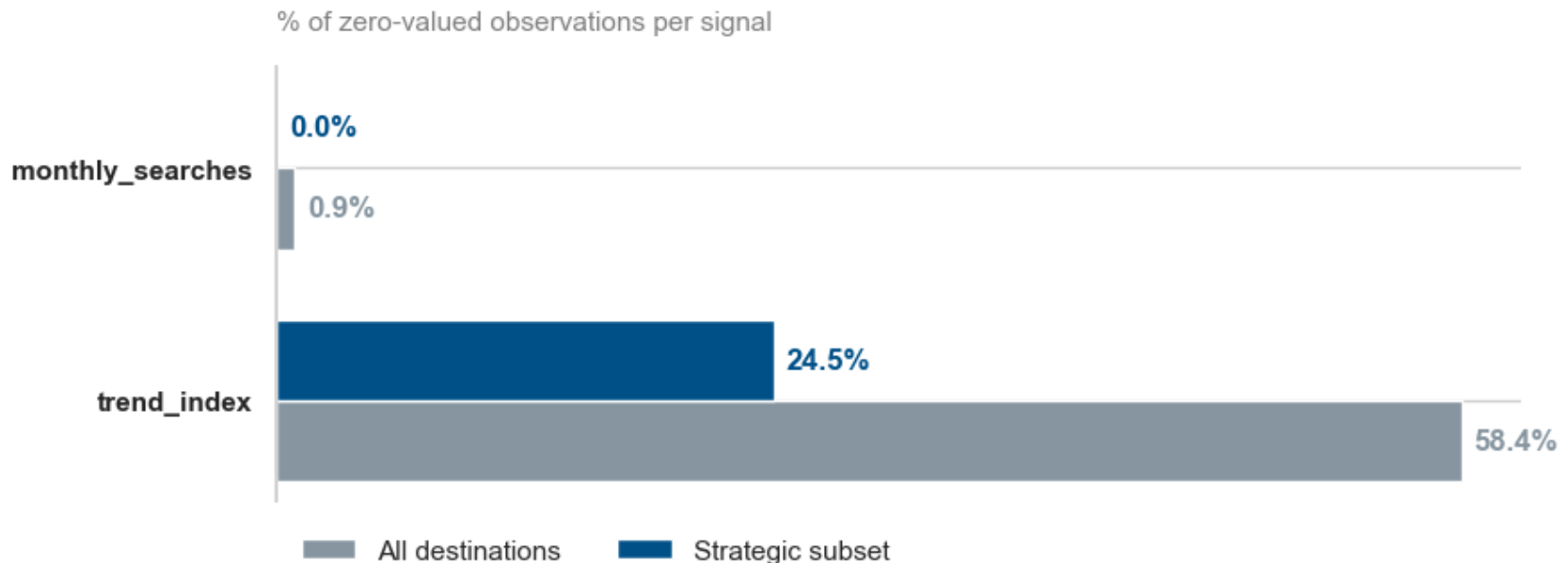
Source: INE ETR 2019–2025

*The subset was designed to reduce demand noise and focus on destinations where digital search behavior is more likely to reflect organized leisure travel; 45 countries +USA (excluded in some analysis).*

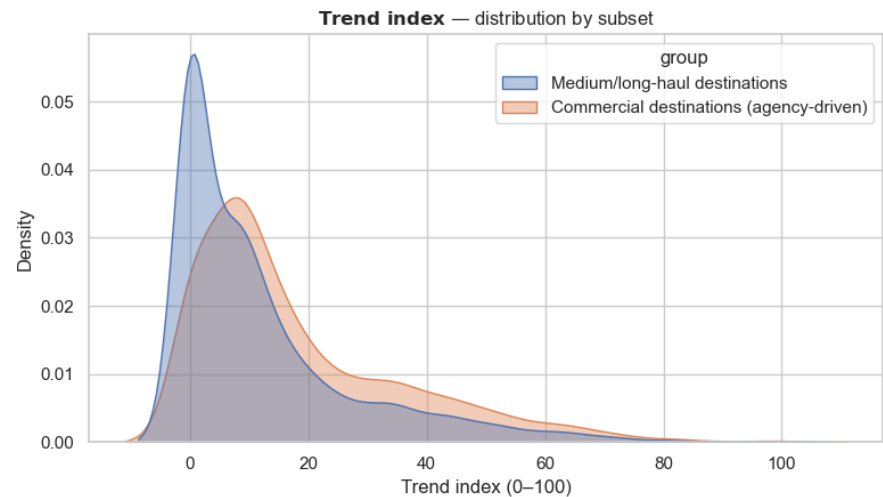
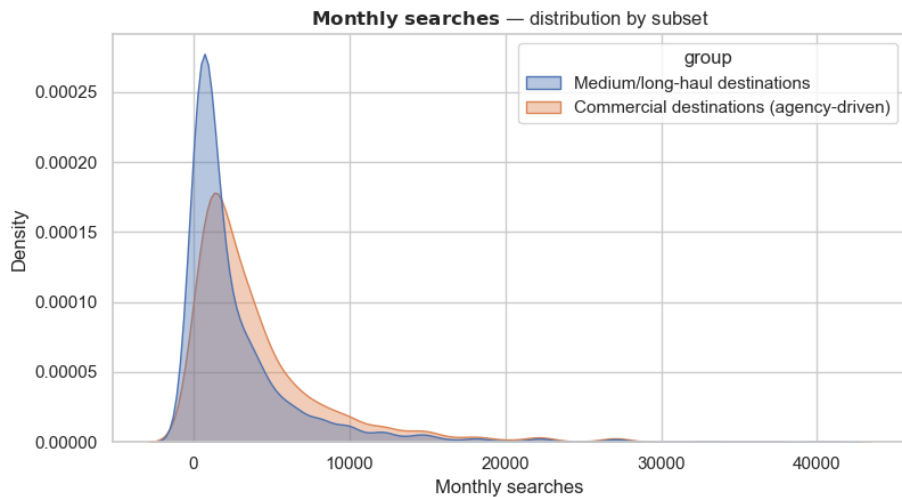
# MONTHLY SEARCHES vs GOOGLE TRENDS

Signal quality: % of zero observations

Filtering for strategic destinations reduces months with missing demand signals



# SIGNAL DISTRIBUTIONS: Strategic filtering improves data quality while preserving market structure



- **Market Alignment between subsets**
- **Noise Reduction:** filter drops zero-search countries.
- **Demand distribution:** concentrated in a few top destinations (long tail).

# NORMALITY TEST JUSTIFIES SPEARMAN

- Shapiro-Wilk Test:

H0: the sample has a Gaussian distribution.

H1: the sample does not have a Gaussian distribution.

| Variable         | Shapiro-Wilk W | p-value     | Result     |
|------------------|----------------|-------------|------------|
| total_tourists   | 0.332          | $\approx 0$ | Not normal |
| monthly_searches | 0.711          | $\approx 0$ | Not normal |
| trend_index      | 0.857          | $\approx 0$ | Not normal |

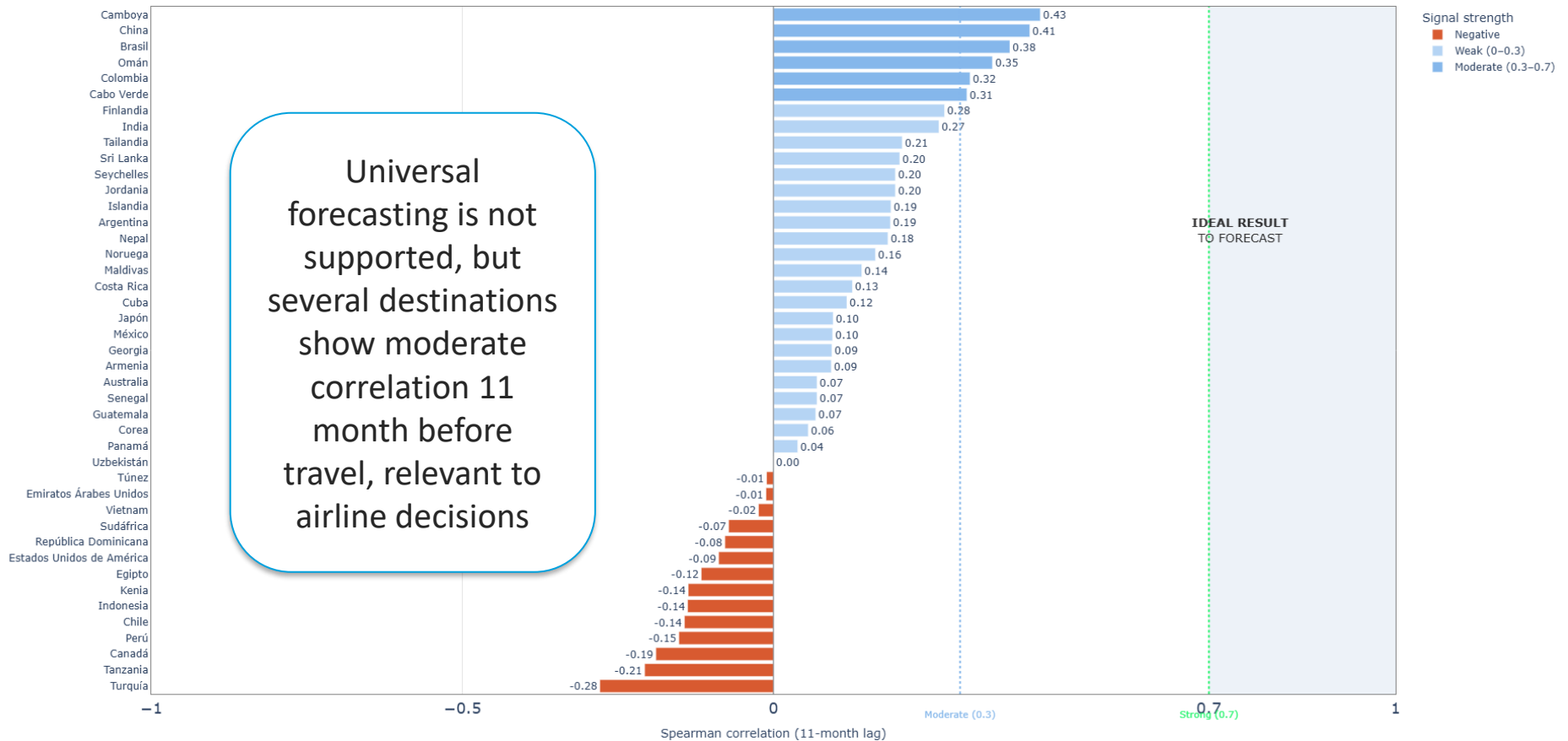
**$p < 0,05$ , H1 accepted**



# CORRELATION RESULTS (Lag 11)

## Google searches (“viajar a ”+destination) vs total trips

### Low correlation at lag 11 across 45 strategic destinations



6 of 45 destinations (13%) showed useful signal at airline planning horizon

# CORRELATION RESULTS (different lags)

Search-demand correlation at strategic lag windows (excl. COVID 2020-2021)



# WHY CORRELATIONS REMAINED LIMITED?



Mobile-phone measurement dataset limitations



INE data not segmented by travel motivations



Simplified keyword architecture ("viaje a + destination"), not capturing all search behavior.



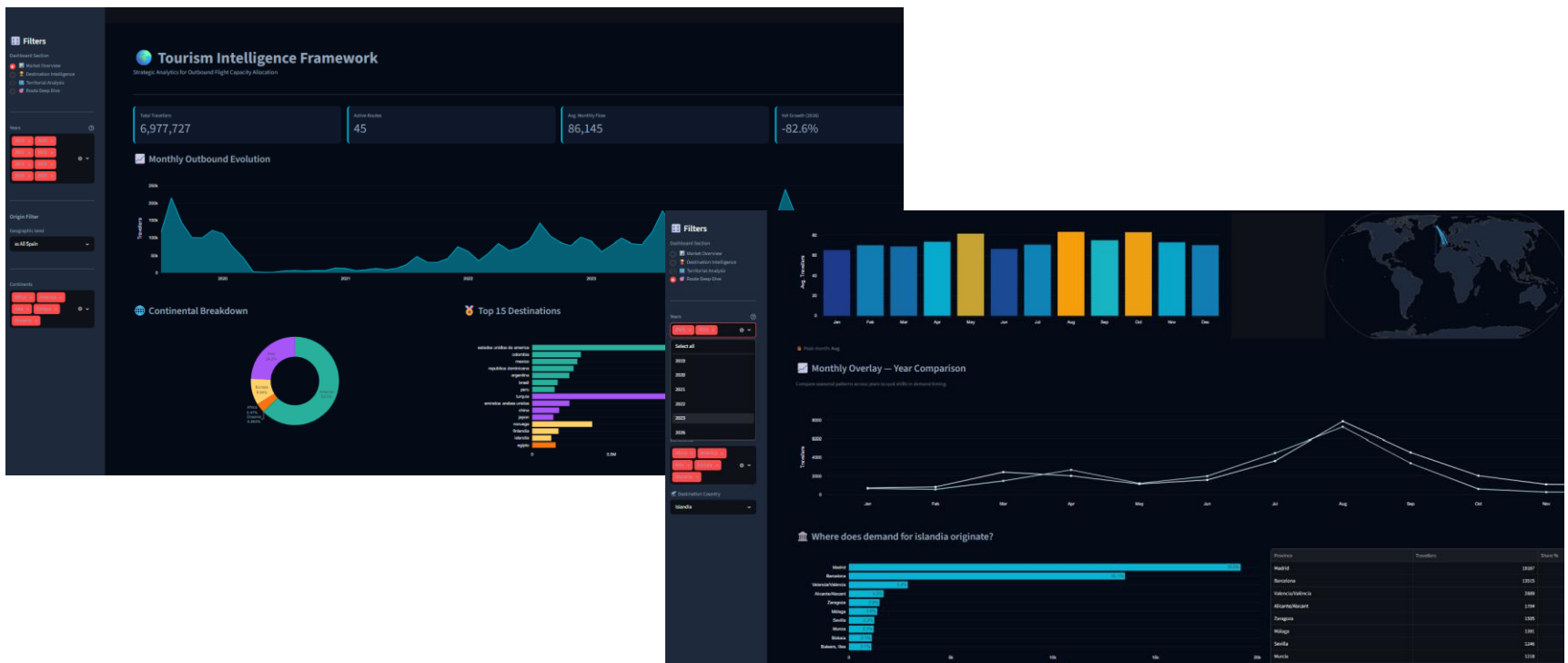
Searches were only analyzed in Spanish



Not all travellers search online

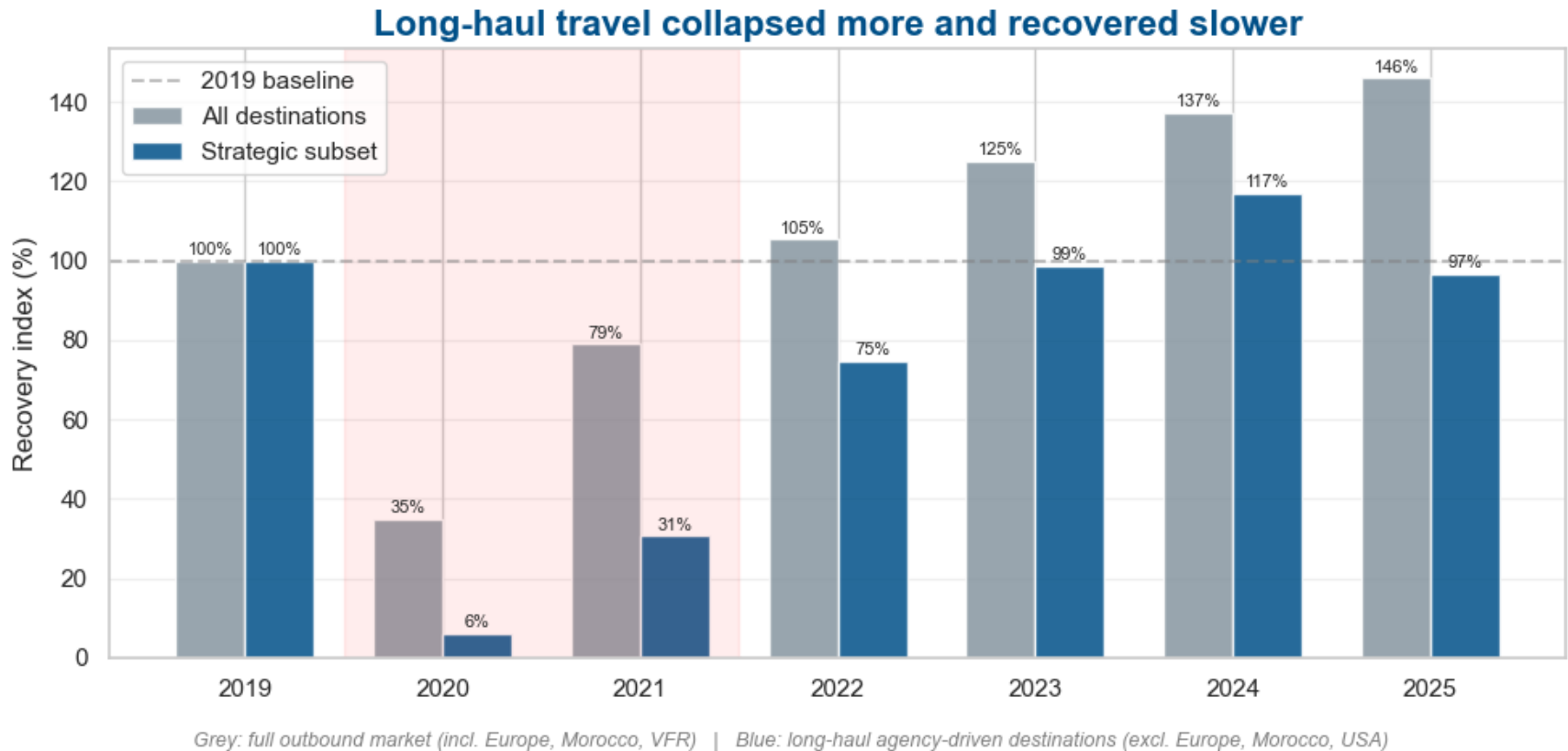
# TOURISM INTELLIGENCE APP (Streamlit)

Useful output to explore related business questions and personalize destination focus analysis.



<https://spain-outbound-tourism-analysis-2026.streamlit.app/>

## Has the client followed the general recovery after COVID?



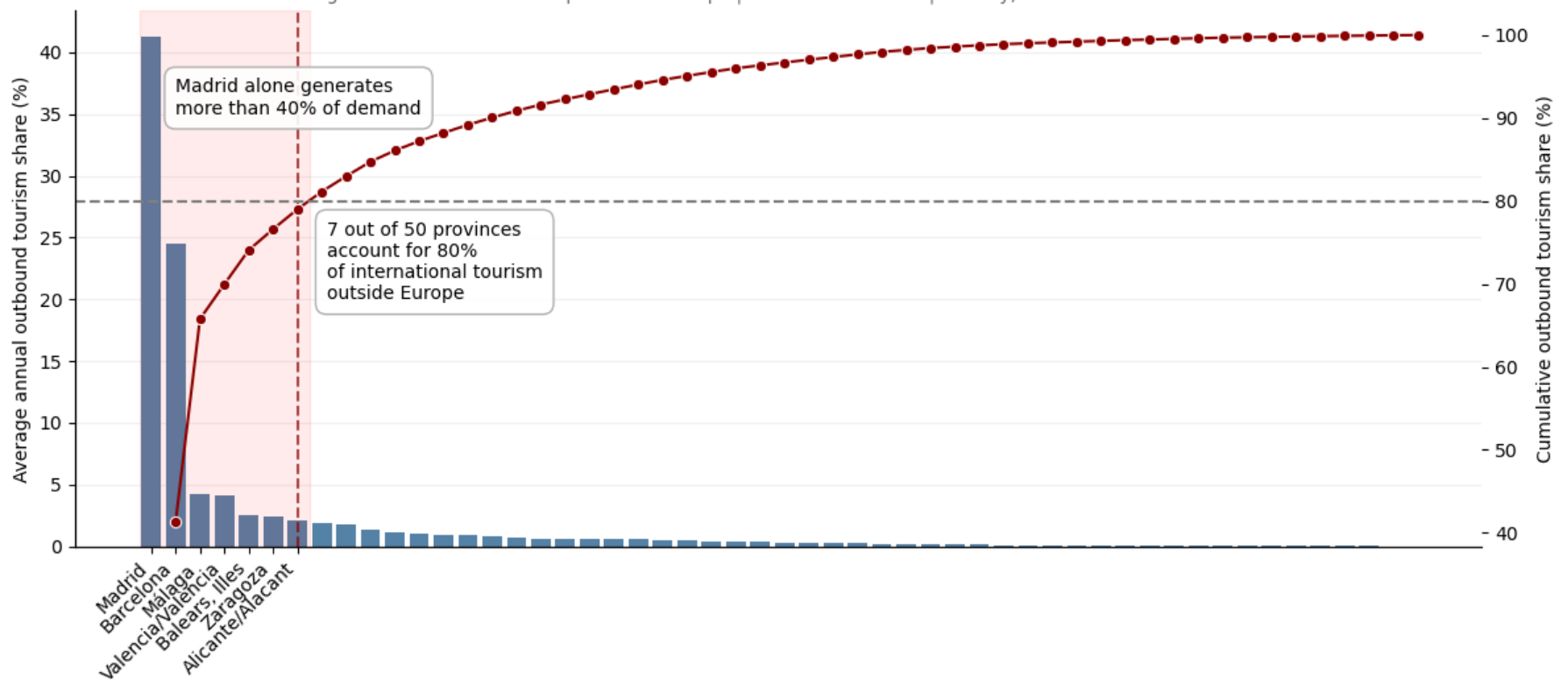
H2 recovery index by segment (Jul-Dec 2019 = 100%)

New provinces where to expand:

Which other provinces generate the highest international outbound demand?

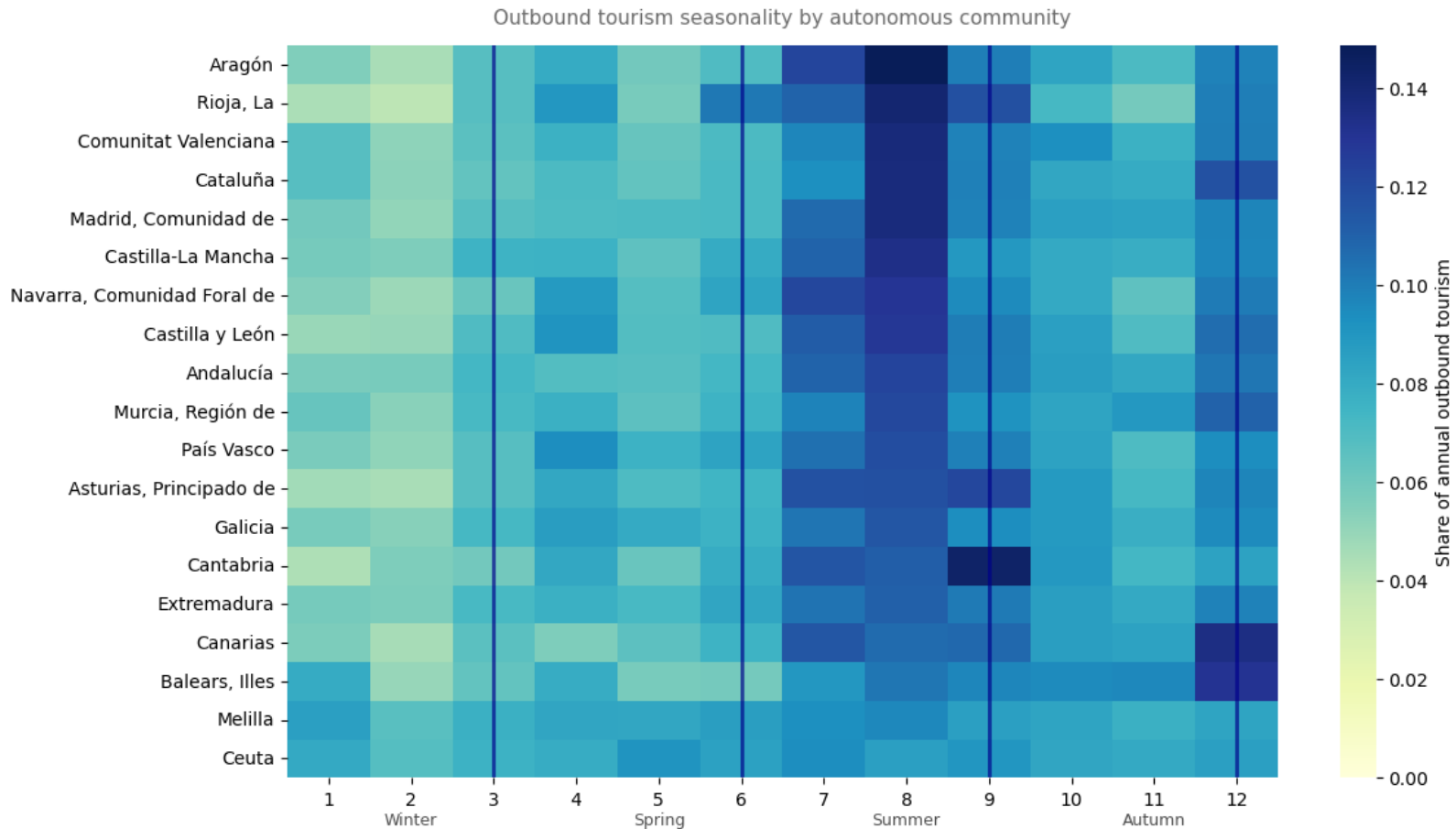
## Spain's outbound tourism outside Europe is highly centralized

Average annual international trips outside Europe | 2019 & 2022-2025 | Norway, Finland and Iceland included



If a destination is successful, what are additional top demand months for each destination, besides August?

## Spanish outbound tourism seasonality varies substantially across regions



## CONCLUSIONS

Can digital demand signals anticipate outbound tourism demand to support better airline seat allocation?

→ **Partially.** Useful for 13% of destinations at annual horizon



- Search signals show selective predictive value



- Google searches outperforms Google Trends.



- Signals work best as real-time monitors (lag 0–3 months)



- Search signals are a **complement**, not a substitute, for INE mobility data in seat allocation decisions.



## NEXT DEVELOPMENT STAGES

From digital demand signals to operational tourism intelligence



### **Improve signal quality**

- Data sources with reason segmentation
- Search-query design
- Additional explanatory variables



### **Build operational forecasting models**

- Autoregressive features
- Macroeconomic indicators
- Transport connectivity



### **Improve agency intelligence tools**

- Interactive dashboards
- Airline capacity integration
- Commercial KPIs

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